

Final Year Project Long Proposal

Interactive Projection Mapping Framework Using Computer Vision and Spatial Tracking

Product Name: SpatialVision

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Revision History

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M Anas Bhatti	Mr. Saad Irfan Khan	24-Sept-2026	Section 1	1.0
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1. Project Stream

[✓] Mobile App

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2. Introduction

Introduction In recent years, the convergence of display technologies and spatial computing has redefined how digital media interacts with the physical world. Traditional visual display systems rely heavily on flat, two-dimensional screens or headsets to present augmented content, restricting user engagement within fixed rectangular boundaries [1]. Spatial Augmented Reality (SAR), widely known as projection mapping, overcomes these physical constraints by utilizing everyday physical objects as dynamic display surfaces [2]. By precise alignment of visual projections over complex 3D surfaces, projection mapping merges digital graphics seamlessly with tangible geometry, unlocking immersive experiences across digital art, retail product showcases, medical visualization, and interactive education.

Despite its growing popularity, conventional projection mapping setups remain largely static and non-interactive. Traditional systems require pre-calculated calibration and manual alignment tailored strictly to fixed projector-object positions [3]. Any slight movement or shift in the physical object breaks the visual alignment, leading to noticeable distortion and visual artifacts. Furthermore, most existing setups lack real-time user responsiveness, treating the viewer merely as a passive observer rather than an active participant. Relying on external handheld controllers or physical touch panels to interact with projected content further disrupts the natural, touchless interaction model that spatial displays inherently promise.

By integrating continuous spatial tracking with real-time pose estimation and touchless gesture interaction, this project advances SAR beyond static installations into adaptive, touchless interactive environments. Eliminating the need for manual re-calibration when objects shift significantly reduces technical overhead in interactive displays. Beyond its immediate application as a functional prototype, this work aligns with broader industry shifts toward ambient computing, spatial human-computer interaction, and touchless display interfaces across interactive retail, smart educational tools, and immersive simulation environments.

3. Project Description

Traditional projection mapping systems are static, requiring manual calibration that breaks whenever an object moves [3]. This project solves that problem by building an

interactive, touchless projection system that adapts visuals to object movement in real time. The main goal is to track physical objects as they shift or rotate and dynamically warp projected graphics so they stay perfectly aligned. To make interaction natural, the system uses computer vision to recognize hand gestures, letting users control projected content without touchscreens or handheld controllers. The final outcome is a working prototype for rigid 3D objects [4] in indoor settings, providing an intuitive interface for interactive retail, smart classrooms, and public exhibits.

The underlying pipeline processes video input using Python and OpenCV to detect surface boundaries and track spatial orientation in 3D space [4]. Estimated pose coordinates (x, y, z position and rotation) feed directly into a rendering pipeline built in Unity, which warps and transforms the graphics before projecting them back onto the object. In parallel, MediaPipe analyzes camera frames to detect hand landmarks and track gestures, translating movements into control commands like swapping projected animations or selecting menu items [5]. A calibrated camera-projector setup closes the feedback loop, continuously matching real-time computer vision tracking with physical projection output.

By combining low-latency tracking algorithms with touchless gesture interaction, this project demonstrates how spatial computing can operate seamlessly without expensive headset hardware or rigid, permanent physical mountings. The system bridges the gap between digital content and physical physical geometry, reducing the setup overhead usually required for interactive spatial displays. Ultimately, this approach establishes a scalable baseline for future spatial augmented reality applications, paving the way for adaptive interfaces in indoor advertising, immersive educational tools, and smart environments.

4. Real World Problem

In modern industries such as retail showcases, museum exhibitions, digital education, and architectural visualization, spatial computing and Spatial Augmented Reality (SAR) are increasingly relied upon to create engaging customer and visitor experiences. Traditional display technologies depend heavily on flat 2D screens or personal headsets like VR/AR goggles. However, flat displays constrain digital media within fixed rectangular borders, while headsets isolate users from one another, preventing shared social interactions. Projection mapping overcomes these physical barriers by transforming everyday physical objects into dynamic display surfaces, merging digital imagery directly onto tangible 3D geometry without requiring users to wear cumbersome hardware [6].

Despite its growing adoption, traditional projection mapping suffers from significant operational limitations that hinder its widespread deployment in real-world environments. Conventional setups rely on pre-calculated geometric alignments tailored strictly to fixed

projector and object positions [3]. If a target object is moved, rotated, or accidentally bumped, the projected content immediately misaligns, leading to noticeable visual distortion and breaking the immersive experience. Re-aligning the projection manually requires specialized technical expertise and substantial downtime, making existing setups impractical for dynamic environments like retail spaces, museum exhibits, or interactive classrooms where physical layout changes occur frequently [7].

Furthermore, most existing projection systems lack real-time user responsiveness, treating the viewer merely as a passive observer. When interaction is offered, it typically depends on physical touch surfaces, touch sensors, or handheld controllers. In public or shared environments, physical contact interfaces raise hygiene concerns, suffer from mechanical wear and tear, and disrupt the natural, touchless interaction model that spatial displays inherently promise. Without continuous spatial tracking and intuitive touchless control, spatial augmented reality remains confined to costly, static installations, limiting user engagement and imposing high technical overhead on organizations.

5. Problem Statement

Spatial Augmented Reality (SAR) has transformed interactive media by projecting visual content directly onto physical 3D surfaces, eliminating the need for headsets or flat screens [1]. However, its effectiveness in dynamic environments—such as retail displays, museums, and educational spaces—is severely restricted by traditional calibration methods. When target objects move, rotate, or shift positions, the projected visuals instantly misalign, resulting in severe visual distortion, broken immersion, and high manual setup overhead for non-technical users.

Conventional projection mapping systems lack automated real-time spatial tracking and touchless user interaction, causing immediate visual alignment failures whenever target objects shift position and forcing reliance on physical controllers or touch panels. This static nature limits spatial displays to rigid, non-interactive installations that require frequent technical intervention. Without continuous computer vision tracking, existing setups fail to maintain real-time alignment or support natural, touchless interaction for dynamic user environments [8].

Failing to address these tracking and interaction bottlenecks restricts SAR from adopting intuitive, ambient computing models across public and commercial domains. Relying on physical touch interfaces or manual recalibration increases operational maintenance, creates hygiene concerns in public spaces, and degrades user experience. Developing a self-aligning, gesture-controlled projection system is essential to bridge the gap between static projection and adaptive, touchless spatial interaction.

6. Names of existing similar FYPs (if repeated)

N/A.

7. New Functionalities (main) to be added Features

N/A.

8. Application Review

Commercial projection mapping software like HeavyM, MadMapper, and Resolume Arena are widely used for visual setups in stage performances, digital art installations, and commercial exhibits. HeavyM and MadMapper focus on geometric warping, grid mapping, and edge blending for visual designers and event technicians. Resolume Arena, on the other hand, is built primarily for live video mixing and performance setups. These tools offer rich visual effect libraries, multi-projector support, and timeline controls that allow operators to manually align digital content onto physical 3D structures with high graphical quality.

Despite their rendering capabilities, current commercial tools struggle in dynamic environments because they rely entirely on static, manual setup. If a physical object shifts, rotates, or gets bumped, the software cannot adjust the graphics on its own. The projected image immediately distorts until a technician manually recalibrates the grid. Additionally, these platforms do not include built-in computer vision for touchless interaction. Users must rely on physical controllers, keyboard shortcuts, or MIDI equipment to change content, creating technical friction for everyday users in interactive retail or educational spaces.

The system developed in this project solves these problems by combining real time spatial tracking and camera based gesture recognition directly into the rendering pipeline. By using continuous camera input to track object movement and hand gestures, the application automatically warps the projected graphics to match physical movement while enabling touchless control.

8.1 SurfaBeam [9]

SurfaBeam Pro is a cross-platform projection mapping app that lets creators warp, mask, and align video onto physical surfaces straight from a mobile device or PC no bulky media servers or expensive setups required.

The key characteristics of the application are listed below.

- **T**ouch-Optimized Real-Time Engine: Leverages on-device GPU processing so you can warp meshes, pin corners, and control timelines directly with touch gestures

without lag.

- **U**niversal Interface Across Devices: Delivers the exact same layout, performance, and workflow whether you run it on iOS, Android, macOS, or Windows
- **1**00% Offline and Private: Runs entirely on local hardware with zero cloud rendering, accounts, or background data tracking.

The limitations of the application are listed below.

- **N**o Automated Cloud Syncing: Projects don't sync between platforms automatically, so files have to be moved manually between devices.
- **G**ated Free Tier: The trial mode adds a watermark, displays a launcher prompt, and locks project saving and exporting behind the Pro license.
- **B**andwidth and Cable Dependency: Output stability relies heavily on a high-quality display adapter (USB-C/HDMI) or a stable 5GHz Wi-Fi link to prevent frame drops during live projection.

8.2 Lazy Lightning [10]

Lazy Lighting (by Audity Apps LLC) is a mobile-first projection mapping application for iOS and Android. It lets creators, VJs, and DIY decorators shape, warp, and align video effects or personal media onto real-world objects directly from a phone or tablet.

The key characteristics of the application are listed below.

- **A**udio and MIDI Reactive Controls: Features over 100 generated live visuals that dynamically sync to tempo, microphone input, or MIDI controllers.
- **C**omprehensive Mapping Toolkit: Includes touch-native corner warping, masks, curved shapes, box mapping, and drawing tools directly on mobile devices.
- **S**tandalone "Lazy Lighting Link" Offloading: Allows users to export and transfer completed shows to dedicated TV/projector receiver apps to play without keeping the phone connected.

The limitations of the application are listed below.

- **S**ingle-Projector Focus: Built specifically for simple single-display setups, lacking multi-projector edge blending and advanced spatial calibration required for large architectural mapping.
- **P**ay walled Feature Set: Key mapping features, show export capabilities, and advanced visual templates are locked behind a paid Pro subscription tier.

8.3 Pro Mapper [11]

Pro Mapper is an iOS-exclusive mobile projection mapping solution that turns any iPhone or iPad into an instant spatial visual studio. Powered by Apple's Metal GPU engine, it allows event creators, visual artists, and performers to warp meshes, trigger audio-reactive effects, and drive interactive 3D portals on physical surfaces without bringing a laptop.

The key characteristics of the application are listed below.

- **A**dvanced Metal GPU Performance: Powered by Apple's Metal rendering engine to deliver real-time, touch-reactive fluid visualizers and low-latency multi-surface warping.
- **I**nteractive FX and Head-Tracking Portals: Includes unique creative capabilities such as head-tracked 3D projection boxes, live person segmentation/glowing outlines, and animated typography.
- **M**ulti-Layer Timeline Control: Features a unified global timeline with individual lanes, clip layering, stroke-by-stroke drawing animations, and direct video export to Photos.connected.

The limitations of the application are listed below.

- **A**pple Ecosystem Exclusivity: Available only on iOS, iPadOS, and macOS (M-series), completely excluding Android and Windows operating systems.
- **S**ingle-Display Output Constraints: Lacks multi-projector edge blending and advanced spatial calibration required for large-scale architectural mapping installations.

8.4 Holo Mapper [12]

A node-based visual programming environment built for custom, highly interactive installations driven by live sensors, cameras, and audio feeds.

The key characteristics of the application are listed below.

- **S**ubscription-Free Flat-Rate Pricing: Available as a one-time purchase on Windows and SteamOS without recurring monthly subscription fees.
- **M**ulti-Format Local Playback: Easily loads and warps video files, animated GIFs, image loops, and direct live capture feeds onto physical targets.
- **L**ightweight Desktop Engine: Designed for fast deployment on standard PCs and handheld SteamOS devices with low CPU/GPU resource overhead.

The limitations of the application are listed below.

- **N**o Multi-Projector Edge Blending: Built for single-display mapping setups, making it unsuitable for complex multi-projector architectural displays.
- **B**asic Toolset: Lacks advanced VJ features like audio-reactive effect generators, DMX/LED lighting integration, and complex timeline sequencing.

8.5 Product: SpatialVision (Proposed)

This project develops a self-aligning projection mapping system using computer vision to track object movements in real time.

Automatically warps projected graphics onto shifted 3D surfaces and detects hand gestures to eliminate manual recalibration while enabling touchless control.

The key characteristics of the application are listed below.

- **R**eal-time Surface and Pose Tracking uses computer vision to continuously monitor 3D object movement and rotation in physical space. This allows the system to dynamically warp projected graphics so that they stay perfectly aligned without manual recalibration.
- **T**ouchless Gesture Control captures user hand movements through camera feedback to translate gestures into direct interactive commands. This enables intuitive, touchless control over projected media without requiring physical screens or handheld hardware.

Table 1 presents a comparative analysis of key features among various applications. The comparison includes crucial functionalities for effective management. The comparison aims to highlight the unique offerings of each application and to emphasize the comprehensive capabilities of the Proposed System.

Table 1: Applications Comparison

Features	**Applications**				
	SurfaBeam [9]	LazyLighting [10]	ProMapper [11]	HoloMapper [12]	SpatialVision
Real-Time 3D Surface Tracking	✓	✗	✗	✗	✓

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Table 1 – *Continued from previous page*

Features	Applications				
	App 1 [9]	App 2 [10]	App 3 [11]	App 4 [12]	Proposed System
Touchless Gesture Control	✓	✗	✗	✗	✓
Wireless Mobile Control App	✓	✓	✓	✓	✓
Self-Aligning Projection Loop	✗	✗	✗	✗	✓
Hygienic Display Interface	✗	✓	✗	✓	✓
Dynamic Preset Switching	✓	✗	✗	✓	✓
Live Mobile Stream Preview	✗	✗	✗	✓	✓
Interactive Projection Mini-Games	✓	✗	✗	✗	✓

8.6 Modules Description

Module 1: Camera & Projector Hardware Interface

- Webcam configuration (resolution, FPS, exposure)
- Projector output and multi-monitor setup
- Camera-projector frame synchronization
- Latency measurement and compensation

Module 2: Mobile Application Interface (Android Native)

- Deliver an Android native app for remote projection management.
- Display live camera previews, system diagnostics, and calibration controls.
- Enable seamless switching between projection presets, scenes, and mini-games.
- Ensure real-time, low-latency wireless communication using WebSockets.

Module 3: Backend and API Integration (FastAPI)

- Handle server-side logic and AI model integration.
- Manage RESTful APIs for system configurations and preset data.
- Use WebSockets for real-time mobile control and live previews.
- Ensure secure local authentication and device authorization.

Module 4: Hand Tracking & Gesture Recognition (OpenCV / MediaPipe)

- HD webcam frame capture and preprocessing

- 21-point 3D hand landmark tracking (MediaPipe)
- Gesture recognition (point, pinch, swipe, palm)
- Gesture-to-projection coordinate mapping

Module 5: Preset & Content Management

- Manage preset visual layouts, interactive scenes, and game templates.
- Handle dynamic content switching and scene transitions.
- Parse scene configuration profiles and custom layout parameters.
- Provide administrative tools for preset creation and content curation.

Module 6: Projection Graphics & Mini-Games

- Real-time scene and UI rendering
- Gesture-driven hover, select, and drag events
- Gesture-reactive mini-games with scoring
- Dynamic scene loading from presets

Module 7: Automatic Projector-Camera Calibration (Homography)

- ArUco marker projection and detection
- Automatic homography matrix computation
- Closed-loop reprojection error correction
- Auto re-calibration on surface movement

Module 8: Usage & Performance Analytics

- Session duration and gesture count tracking
- FPS, latency, and CPU/GPU monitoring
- Tracking accuracy and calibration error logs
- Exportable CSV/JSON reports

Module 9: Real-Time Communication

- REST API endpoints for system control and status updates
- Low-latency WebSocket communication for real-time data exchange
- MJPEG video streaming for live mobile previews
- Heartbeat and automatic reconnection for stable communication

Module 10: Surface Warping & Distortion Correction

- Perspective warping using homography
- Mesh-based keystone distortion correction

- Edge blending and surface masking
- Real-time warp update on surface movement

Module 11: Security & Network Access Control

- Pairing code and JWT authentication
- Role-based access (admin/viewer)
- LAN-only access for authorized devices
- Local camera processing with no cloud upload

Module 12: Preset Storage & Asset Management System

- Store and manage projection assets, game settings, and surface coordinates.
- Quickly load visual presets and calibration data from the local database.
- Save manual calibration changes even after restarting the system.
- Use structured JSON data to keep system information synchronized.

9. Modules

Table 2 provides a breakdown of project modules and assigned students based on deadlines, showcasing their roles and responsibilities.

Table 2: Module Distribution Among Students

Deadline	Modules	Student
P1 Mid	Camera & Projector Hardware Interface	Zahra Tul Hussan Syed (23108195)
	Mobile Application Interface (Android Native)	M Anas Bhatti (23108243)
	Backend and API Integration (FastAPI)	Syed Ali Tarikh (23108189)
P1 Final	Hand Tracking & Gesture Recognition (OpenCV / MediaPipe)	Zahra Tul Hussan Syed (23108195)
	Preset & Content Management	Syed Ali Tarikh (23108189)
	Projection Graphics & Mini-Games	M Anas Bhatti (23108243)
P2 Mid	Automatic Projector-Camera Calibration (Homography)	M Anas Bhatti (23108243)
	Usage & Performance Analytics	Zahra Tul Hussan Syed (23108195)
	Real-Time Communication (FastAPI / WebSocket)	Syed Ali Tarikh (23108189)

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Table 2 – *Continued from previous page*

Deadline	Modules	Student
P2 Final	Surface Warping & Distortion Correction	Zahra Tul Hussan Syed (23108195)
	Security & Network Access Control	Syed Ali Tarikh (23108189)
	preset Storage & Asset Management System	M Anas Bhatti (23108243)

10. Development Environment

Tools:

- Python: Backend system logic, computer vision pipelines, and system orchestration.
- Android Studio & Kotlin: Native Android management app for mobile control..
- OpenCV: Real-time video processing, feature tracking, and homography matrix calculations.
- MediaPipe: Machine learning pipeline for 21 3D hand landmark tracking and gesture recognition.
- VS Code / Android Studio: Primary IDEs for coding, app building, and debugging.
- Git & GitHub: Version control system and source code repository management.

DBMS:

- Firebase Firestore / SQLite: Real-time database for storing system profiles, calibration settings, and layout presets.

Platform:

- Windows 11 / Linux (Ubuntu): Host OS environment running real-time vision algorithms and projection pipelines.
- WebSockets / REST API: Networking protocols for real-time control between the mobile app and Python backend.

11. Significance

Traditional Spatial Augmented Reality (SAR) systems often depend on fixed geometries and manual alignment, making them difficult to use in dynamic environments. This project addresses these limitations by introducing a self-aligning and touchless projection framework. In environments such as museums, retail spaces, and interactive classrooms, objects may be moved or rearranged frequently. Automated recalibration and hand-gesture interaction can reduce manual effort while providing a more flexible and hygienic user experience.

The project combines real-time computer vision-based pose tracking, perspective transformation, and MediaPipe gesture detection into a low-latency control pipeline. A mobile control application further allows users to manage calibration and display content wirelessly without requiring specialized hardware or advanced technical skills. This can reduce maintenance requirements and make the system easier to operate.

By using affordable webcams, standard projectors, and open-source software, the proposed framework aims to provide a scalable and cost-effective solution for spatial computing. It supports the growing use of touchless interaction and can make interactive projection experiences more accessible in educational, commercial, and public indoor environments.

12. Tools and Techniques

The core implementation of this project combines real-time computer vision algorithms, machine learning frameworks, dynamic rendering engines, and a dedicated mobile management application to achieve an adaptive projection mapping system. Spatial surface tracking relies on OpenCV for feature detection, pose estimation, and homography matrix calculations that warp digital graphics to match shifting 3D physical surfaces in real time. Touchless user interaction is handled by Google's MediaPipe framework, which detects 21 3D hand keypoints from incoming video feeds to translate hand gestures into interactive control triggers. A cross-platform mobile application built with Flutter communicates with a central Python backend over Wi-Fi via WebSocket protocols, allowing users to remotely toggle camera detection feeds, adjust projector calibration parameters, and manage content navigation. The complete pipeline runs on a central PC workstation connected to a high-frame-rate HD webcam, a digital projector, and rigid target display objects operating in a controlled indoor setup.

- Mobile App Framework: Android Native (Kotlin/Java) for building a dedicated mobile management app that remotely controls camera feeds, projector calibration settings, and system navigation.
- Network and Communication Protocols: WebSockets and RESTful APIs implemented using FastAPI in Python to enable low-latency, real-time wireless communication between the mobile app and the PC processing engine
- Programming Languages: Python for backend system orchestration, computer vision processing, and matrix transformations; Dart for mobile application development.
- Computer Vision Framework: OpenCV (opencv-python) to manage live camera streams, feature detection, pose estimation, and 3×3 homography perspective warping.

- Gesture Recognition Pipeline: MediaPipe Hands to extract 21 3D hand landmark coordinates for touchless gesture detection (swiping, tapping, selecting) without physical sensors.
- Hardware Configuration: A standard 1080p USB webcam (30–60 FPS), an HD digital projector, a host PC workstation, a mobile smartphone, and rigid 3D geometric objects in a controlled indoor space.

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